Meta-Guided Action-Manifold Smoothing for Secure Ackermann Vehicle Platoons under Hybrid Attacks

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***Abstract*—With the increasing integration of vehicle-to-everything (V2X) communication and autonomous driving, connected and automated vehicle (CAV) platoons have become an important platform for cooperative transportation. However, hybrid denial-of-service (DoS) and false-data-injection (FDI) attacks can corrupt exchanged observations and control commands, causing the reward signals used by reinforcement-learning-based controllers to deviate from physical maneuver objectives.**

To address this problem, a Meta-Guided Action-Manifold Smoothing (Meta-AMS) framework is proposed for secure cooperative maneuver learning of Ackermann vehicle platoons under hybrid DoS–FDI attacks. A physics-aware Meta Reward Teacher exploits trusted physical information during centralized training to correct attack-induced reward misalignment. An auxiliary action-value model evaluates feasible neighborhoods of joint speed–steering actions and supplies an additional advantage signal to Heterogeneous-Agent Proximal Policy Optimization (HAPPO), thereby introducing local action-value regularization into policy updates.

Furthermore, a cooperative lane-change problem is formulated under the front-wheel-steering constraint of Ackermann vehicles. A progressive hybrid-attack evaluation protocol and a strict whole-platoon completion criterion are established to assess maneuvering performance under increasingly severe attacks. Theoretical analysis characterizes the teacher meta-gradient relationship, the local action-value regularization induced by neighborhood averaging, and the conditional tracking boundedness of the learned platoon controller under admissible disturbances.

Simulation results show that the verified Meta-AMS policy achieves a whole-platoon completion rate of 1.00 through the hard-a attack stage and 0.75 at hard-b, with zero recorded collision and road-termination rates across the reported evaluation matrix. Comparisons with MARL baselines further characterize its performance under the same progressive attack protocol.

***Index Terms*—Vehicle platooning, Ackermann steering, hybrid cyber attacks, multi-agent reinforcement learning, reward shaping, action-manifold smoothing, HAPPO, safety-critical control.**

I. Introduction

With the development of autonomous driving and vehicle-to-everything (V2X) communication, connected and automated vehicle (CAV) platoons have become an important component of intelligent transportation systems. By exchanging vehicle states and coordination information, platooning systems can improve traffic efficiency and enable coordinated vehicle motion. However, dependence on wireless communication exposes vehicle platoons to cyber threats that may compromise the information required for safe and effective coordination [1], [2].

Among these threats, denial-of-service (DoS) and false-data-injection (FDI) attacks are particularly challenging. DoS attacks impair information availability through packet losses or communication interruptions, whereas FDI attacks compromise data integrity by manipulating transmitted observations or control commands. Their combined effects can degrade vehicle coordination and distort the feedback used to evaluate control performance. Existing security-control methods, including resilient estimation, fault-tolerant control, and game-theoretic approaches, mitigate attack effects [3], [4]. Their reliance on system models, specified attack structures, or explicit adversarial assumptions limits direct application to nonlinear cooperative maneuvering tasks.

Multi-agent reinforcement learning (MARL) offers an alternative for coordinating multiple vehicles through centralized training and decentralized execution (CTDE) [5]. Algorithms such as Multi-Agent Proximal Policy Optimization (MAPPO) and Heterogeneous-Agent Proximal Policy Optimization (HAPPO) provide established policy-optimization mechanisms for cooperative decision-making [6], [7]. Under hybrid DoS–FDI attacks, corrupted observations and commands distort the reward signals used during training. The resulting reward misalignment can drive policy optimization away from the physical maneuver objective, even when the learning procedure remains numerically stable.

Meta-gradient reward shaping addresses this issue by adapting the learning reward according to a trusted physical-performance objective. In the reference MGRS framework, a physics-aware Reward Teacher exploits physical states and executed inputs available during centralized training to adjust the reward-shaping signal [8], [9]. Reward correction aligns the learning objective, while local action-value estimates remain subject to variation during policy updates. This distinction is relevant to Ackermann vehicles, whose longitudinal speed and front-wheel steering jointly determine vehicle curvature, lateral motion, and inter-vehicle spacing. Reliable cooperative maneuver learning therefore requires both physically consistent reward correction and evaluation of nearby feasible actions [10].

Cooperative lane changing introduces motion constraints that longitudinal car-following alone does not represent. For an Ackermann vehicle, the rear wheels provide driving motion while the front wheels steer, and the resulting nonholonomic motion couples longitudinal progression with lateral displacement and heading evolution [11]. Under hybrid attacks, differences in vehicle responses can cause platoon members to enter

and leave the lane-change segment at different times. The control problem therefore requires coordinated path following and spacing regulation for the entire platoon; average performance or partial maneuver completion is insufficient [12].

To address these challenges, this paper proposes a Meta-Guided Action-Manifold Smoothing (Meta-AMS) framework for secure cooperative lane-change learning of Ackermann vehicle platoons under hybrid DoS–FDI attacks. The framework retains physics-aware meta-gradient reward correction and incorporates action-manifold smoothing into the HAPPO update through feasible neighborhoods of joint speed–steering actions [13]. The two mechanisms address complementary aspects of learning: the Meta Reward Teacher adjusts the optimization objective using trusted physical performance, whereas the AMS critic provides local action-value information for policy updates. The resulting method is evaluated using a progressive hybrid-attack protocol and a strict whole-platoon completion criterion.

The main contributions of this paper are summarized as follows.

- A Meta-AMS learning framework is developed by integrating physics-consistent reward correction with local action-value regularization for vehicle platoons under hybrid DoS–FDI attacks. The Meta Reward Teacher aligns reward adaptation with trusted physical performance during centralized training, while feasible-neighborhood value evaluation supplies an auxiliary advantage signal to the HAPPO update.
- A secure cooperative lane-change problem is formulated for a five-vehicle Ackermann platoon under front-wheel-steering constraints. The evaluation accounts for path tracking, heading evolution, inter-vehicle spacing, and whole-platoon completion under progressively intensified hybrid attacks.
- Theoretical properties of the proposed learning framework are analyzed, including the teacher meta-gradient relationship, the local action-value regularization induced by symmetric neighborhood sampling, and conditional tracking boundedness under admissible disturbances. Simulation studies assess the resulting maneuvering performance and compare Meta-AMS with MARL baselines under a common progressive attack protocol.

II. Problem Formulation

A. Five-Vehicle Ackermann Platoon

Consider one leader and four followers, indexed by $i \in \{1,\dots,5\}$ for the five physical vehicles. The local planar state is

$$\boldsymbol{x}_i = [x_i, y_i, \psi_i, v_i]^{\mathsf{T}}, \tag{1}$$

where (x_i, y_i) is the body reference position, ψ_i is yaw, and v_i is the forward speed. With wheelbase L and front steering angle δ_i, the kinematic core follows the standard Ackermann bicycle abstraction [11]:

$$\dot{x}_i = v_i\cos\psi_i, \quad \dot{y}_i = v_i\sin\psi_i, \quad \dot{\psi}_i = \frac{v_i}{L}\tan\delta_i, \quad \dot{v}_i = a_i + w_i \tag{2}$$

The rear steering angle is identically zero. The policy action is a normalized pair

$$\boldsymbol{a}_i = [a_i^v, a_i^\delta]^{\mathsf{T}} \in [-1,1]^2, \tag{3}$$

which is mapped to center speed and center steering by $v_i = v_{\max} a_i^v$ and $\delta_i = \delta_{\max} a_i^\delta$. The implementation uses $L = 0.14325$ m, $v_{\max} = 1.20$ m/s, and $\delta_{\max} = 0.32$ rad.

For track width T, the two front steering targets are generated from the center steering command:

$$\delta_{i,L} = \tan^{-1}\left(\frac{L\tan\delta_i}{L - \frac{T}{2}\tan\delta_i}\right), \tag{4}$$

$$\delta_{i,R} = \tan^{-1}\left(\frac{L\tan\delta_i}{L + \frac{T}{2}\tan\delta_i}\right). \tag{5}$$

The rear axle receives only wheel angular-velocity targets. Thus, the action interface itself enforces the front-steering-only Ackermann structure.

B. Shared Spatial Lane-Change Path

All five vehicles follow one geometric reference. The reference is a quintic spatial transition. For a vehicle-local longitudinal coordinate x, define

$$\zeta(x) = \mathrm{clip}\left(\frac{x - x_s}{\ell}, 0, 1\right), \qquad h(\zeta) = 10\zeta^3 - 15\zeta^4 + 6\zeta^5, \tag{6}$$

and

$$y_{\mathrm{ref}}(x) = y_s + (y_e - y_s)h(\zeta(x)). \tag{7}$$

The verified protocol uses $x_s = -7.75$ m, $\ell = 6.0$ m, $y_s = -0.065$ m, and $y_e = -1.160$ m. The five initial longitudinal positions are separated by 1.50 m. The reference provides a common centerline and heading target, while each follower additionally receives predecessor-relative features.

The local observation contains predecessor-relative position and velocity, ego velocity and yaw rate, centerline and heading error, relative heading, and a vehicle-role identifier. The centralized critic receives the joint observation, while each actor uses only its local attacked observation. This is a centralized-training/decentralized-execution (CTDE) design [5]–[7].

C. Hybrid DoS–FDI Attack

Let $\boldsymbol{o}_i$ denote the clean local observation and $\boldsymbol{u}_i$ the raw command. A general hybrid attack interface can represent FDI vectors and DoS probabilities as

$$\boldsymbol{\xi}_i = G_{\theta_G}(\boldsymbol{z}_i, \boldsymbol{o}_i) = [\boldsymbol{f}_{p,i}, q_{p,i}, \boldsymbol{f}_{a,i}, q_{a,i}]. \tag{8}$$

The attacked observation and command are

$$\widetilde{\boldsymbol{o}}_i = \boldsymbol{\beta}_{p,i} \odot \boldsymbol{o}_i + \boldsymbol{f}_{p,i} + (\mathbf{1} - \boldsymbol{\beta}_{p,i}) \odot \boldsymbol{o}_{i,\mathrm{hold}}, \tag{9}$$

$$\widetilde{\boldsymbol{u}}_i = \beta_{a,i}\boldsymbol{u}_i + \boldsymbol{f}_{a,i} + (1 - \beta_{a,i})\boldsymbol{u}_{i,\mathrm{hold}}, \tag{10}$$

where the binary masks are sampled from the selected attack process. The reported matrices use the bounded `profile` attack mode, with stage-specific FDI and DoS budgets sampled from the fixed evaluation protocol. The CAGAN/generator mode is outside the reported experiments. The evaluation

ladder increases the observation FDI, command FDI, and DoS budgets from *off* through *light*, *easy*, *medium*, and *hard*. At hard-b, the protocol uses position FDI 4.0, acceleration FDI 1.3, and DoS rate 0.18.

D. Strict Whole-Platoon Completion

The maneuver is considered complete only when every vehicle reaches at least 99% spatial progress, with centerline error at most 0.12 m, heading error at most 0.12 rad, predecessor gap error at most 0.35 m, pair lateral error at most 0.20 m, pair distance at least 0.30 m, and no road-boundary exceedance. The conditions must persist for a 2.0 s dwell. This whole-platoon criterion is stricter than a mean spacing score and is motivated by string-stability concerns in platoons [12]. Whole-platoon completion rate is the primary maneuver metric.

III. Meta-Guided Action-Manifold Smoothing

A. Bi-Level Task Direction

Let θ denote the decentralized policy parameters and ϕ the reward-teacher parameters. The attacked local reward is augmented by a learned shaping term:

$$R_\phi(\widetilde{\boldsymbol{o}}_i, \boldsymbol{u}_i) = R_{\text{env}}(\widetilde{\boldsymbol{o}}_i, \boldsymbol{u}_i) + F_\phi(\widetilde{\boldsymbol{o}}_i, \boldsymbol{u}_i). \tag{11}$$

During centralized training, a trusted physical score is computed from true states and executed commands:

$$\mathcal{L}_{\text{phy}} = \frac{1}{5}\sum_{i=1}^{5}\left(\lambda_p e_{p,i}^2 + \lambda_v e_{v,i}^2 + \lambda_a e_{a,i}^2 + \lambda_c e_{c,i}^2 + \lambda_\psi e_{\psi,i}^2 + \lambda_j a_i^2 + \lambda_u \|\boldsymbol{u}_i\|^2\right) + \lambda_{\text{col}} I_{\text{col}}. \tag{12}$$

The ideal lower-level and upper-level problems are

$$\theta^*(\phi) \in \arg\max_\theta J_{\text{in}}(\theta;\phi), \tag{13}$$

$$\min_\phi \quad J_{\text{out}}(\phi) = J_{\text{phy}}(\theta^*(\phi)). \tag{14}$$

This preserves the central idea of meta-gradient reward shaping [8], [9] and the safety-oriented view that performance constraints should be represented explicitly [10]: the teacher is updated from trusted post-update physics, while deployment uses only the learned students.

For a one-step differentiable surrogate,

$$\bar{\theta}^+ = \theta^- + \alpha g_{\text{in}}(\theta^-, \phi), \tag{15}$$

the first-order outer direction is

$$\nabla_\phi J_{\text{out}} = \left(\frac{\partial \bar{\theta}^+}{\partial \phi}\right)^{\mathsf{T}} \nabla_{\bar{\theta}^+} J_{\text{phy}}. \tag{16}$$

The implementation uses a stable proxy based on normalized physical-score changes and the covariance between centered physical advantages and centered teacher outputs. A warm-up and ramp prevent the teacher from dominating before the retention-initialized policy has learned the basic maneuver.

B. Action-Manifold Critic

The AMS module receives a centralized observation $\boldsymbol{s}$ and a joint action $\boldsymbol{a} \in \mathbb{R}^{10}$ formed by five speed–steering pairs. It learns two auxiliary action-value functions Q_1 and Q_2. Let $\{\boldsymbol{d}_j\}_{j=1}^K$ be orthonormal directions obtained from a QR factorization and let $\epsilon = r\ell_a$, where r is the neighborhood radius and ℓ_a is the normalized action limit. The feasible neighbors are

$$\boldsymbol{a}_j^{\pm} = \Pi_{\mathcal{A}_{\text{Ack}}}(\boldsymbol{a} \pm \epsilon \boldsymbol{d}_j), \qquad j = 1, \ldots, K. \tag{17}$$

The projection $\Pi_{\mathcal{A}_{\text{Ack}}}$ clips normalized speed and steering, limits curvature, and preserves the speed–steering pair structure. The smoothed target value is

$$\overline{Q}^{\,t}(\boldsymbol{s}, \boldsymbol{a}) = \frac{1}{2K}\sum_{j=1}^{K} \min_{q\in\{1,2\}} \left[Q_q^t(\boldsymbol{s}, \boldsymbol{a}_j^+) + Q_q^t(\boldsymbol{s}, \boldsymbol{a}_j^-)\right], \tag{18}$$

and the clipped temporal-difference target is

$$y = r + \gamma m\, \overline{Q}^{\,t}(\boldsymbol{s}', \boldsymbol{a}'), \tag{19}$$

where m is the continuation mask. Each critic is trained with a Huber loss against y, followed by a Polyak target update.

The action anchor is the physically executed action, rather than a relabeled projection of the stored transition. The local AMS advantage for the actor update is

$$A_{\text{AMS}} = \min(Q_1, Q_2)(\boldsymbol{s}, \boldsymbol{a}_{\text{exec}}) - \overline{Q}(\boldsymbol{s}, \Pi_{\mathcal{A}_{\text{Ack}}}(\boldsymbol{a}_\pi)) - \rho\sqrt{\text{Var}(Q_{\text{neigh}})}, \tag{20}$$

where $\rho \geq 0$ is an optional local-risk coefficient. The current verified configuration uses $K = 8$, radius $r = 0.25$, Huber parameter 0.3, target smoothing $\tau = 0.01$, and $\rho = 0$.

C. HAPPO Update and Warm-Up

Let A_{HAPPO} be the normalized sequential-policy advantage. The actor receives

$$A_{\text{total}} = \text{Norm}\left(A_{\text{HAPPO}} + \lambda_k \,\text{Norm}(A_{\text{AMS}})\right), \tag{21}$$

where

$$\lambda_k = \begin{cases} 0, & k \leq k_{\text{warm}}, \\ \lambda_{\max} \min\left(\dfrac{k - k_{\text{warm}}}{k_{\text{ramp}}}, 1\right), & k > k_{\text{warm}}. \end{cases} \tag{22}$$

The runner defaults are $\lambda_{\max} = 0.10$, 50 warm-up updates, and a 100-update ramp, but the verified matrix launcher overrides these values to $\lambda_{\max} = 0.02$, 5 warm-up updates, and a 15-update ramp for the reported short stage budgets. Meta-AMS therefore adds a bounded local action signal to the trust-region-style policy update while retaining the task reward and HAPPO objective [6], [14].

D. Training Algorithm

Algorithm 1 summarizes the complete sequence. The attack curriculum and the expert-retention warm start are part of the algorithm, because abruptly enabling the hard attack before the five vehicles can perform the clean spatial maneuver confounds security learning with basic path acquisition.

Algorithm 1 Meta-AMS training under progressive hybrid attacks

Require: BC/retention initialization θ, teacher ϕ, AMS critics Q_1, Q_2, attack profile (or optional generator G)

for attack stages $\ell \in \{\text{off}, \text{light}, \ldots, \text{hard-b}\}$ **do**
  Set the stage attack budgets and reset the strict completion gate
  for student update $k = 1, \ldots, K_\ell$ **do**
    Roll out five Ackermann vehicles with attacked local observations
    Store raw actions and physically executed actions
    Compute attacked rewards and trusted physical teacher scores
    Update the HAPPO students with the retention-aware shaped reward
    Sample feasible AMS neighbors and update twin action-value critics
    Add the warm-started AMS advantage using (21)
    if teacher boundary is active **then**
      Update ϕ using the normalized physical-score/teacher-output proxy
    If generator mode is enabled, update G and its discriminator; otherwise retain the fixed profile sampler
  Evaluate clean and attacked environments with fixed evaluation seeds
  if all completion, dwell, collision, and road gates pass **then**
    Promote the checkpoint to the next attack stage
  else
    Keep the last passing checkpoint and record the failure diagnostics

IV. Theoretical Analysis of Meta-AMS Learning Framework

This section provides theoretical interpretations of the two learning components in Meta-AMS. The Meta Reward Teacher establishes the consistency between the attack-corrupted learning objective and the trusted physical maneuver objective through a bilevel optimization relationship. The AMS module further characterizes the local regularization property of action-value evaluation in the coupled Ackermann action space. Based on the learned policy, the conditional tracking boundedness of the vehicle platoon is finally analyzed under bounded hybrid disturbances.

A. Objective Consistency of Physics-Aware Meta Reward Teacher

Let the five-vehicle platoon consist of N decentralized policies, and collect their parameters as

$$\theta = \mathrm{col}(\theta_1, \ldots, \theta_N). \tag{23}$$

The attacked local reward is represented by

$$R_\phi(\widetilde{\boldsymbol{o}}, \boldsymbol{u}) = R_{\mathrm{env}}(\widetilde{\boldsymbol{o}}, \boldsymbol{u}) + F_\phi(\widetilde{\boldsymbol{o}}, \boldsymbol{u}). \tag{24}$$

The inner objective used by the decentralized policies is

$$J_{\mathrm{in}}(\theta, \phi) = \mathbb{E}_{\widetilde{\tau}} \left[\sum_{t=0}^{T-1} \gamma^t R_\phi(\widetilde{\boldsymbol{o}}_t, \boldsymbol{u}_t) \right]. \tag{25}$$

The trusted physical objective is defined from the physical trajectory and executed commands as

$$J_{\mathrm{phy}}(\theta) = \mathbb{E}_{\tau(\theta)} \Bigg[\sum_{t=0}^{T-1} \Big(\lambda_p \left\| \boldsymbol{e}_{p,t} \right\|^2 + \lambda_s \left\| \boldsymbol{e}_{s,t} \right\|^2 + \lambda_\psi \left\| \boldsymbol{e}_{\psi,t} \right\|^2 + \lambda_v \left\| \boldsymbol{e}_{v,t} \right\| + \lambda_{\Delta u} \left\| \Delta \boldsymbol{u}_t \right\|^2 + \lambda_{\mathrm{col}} I_{\mathrm{col},t} \Big) \Bigg], \tag{26}$$

where the terms measure path-tracking, spacing, heading, velocity, control-smoothness, and collision performance, respectively. The differentiable post-update surrogate is written as

$$\begin{aligned} &\theta^\star(\phi) \in \arg\max_\theta J_{\mathrm{in}}(\theta, \phi), \\ &\bar{\theta}^+(\phi) = \mathcal{U}(\theta, \phi), \qquad J_{\mathrm{out}}(\phi) = J_{\mathrm{phy}}\big(\bar{\theta}^+(\phi)\big). \end{aligned} \tag{27}$$

The reward-teacher mapping F_ϕ compensates for the inconsistency between attacked learning signals and physical cooperative maneuver requirements. The teacher participates only in centralized training, where it determines the optimization direction of the student policies; deployment uses the learned decentralized policies.

Assumption 1 (differentiability of the teacher–policy update). The mapping F_ϕ and the HAPPO update operator $\mathcal{U}$ are differentiable in their respective parameters, and the derivatives required below are integrable so that differentiation and expectation can be interchanged.

For the one-step differentiable post-update policy, write

$$\bar{\theta}^+ = \theta^- + \alpha g_{\mathrm{in}}(\theta^-, \phi), \tag{28}$$

where g_{in} is the inner-policy update direction induced by J_{in}. The following theorem characterizes how the trusted physical objective enters the teacher meta-gradient through the differentiable post-update policy.

Theorem 1 (Meta Reward Teacher Objective Alignment). Under Assumption 1, the outer objective satisfies

$$\nabla_\phi J_{\mathrm{out}} = \left(\frac{\partial \bar{\theta}^+}{\partial \phi} \right)^{\mathsf{T}} \nabla_\theta J_{\mathrm{phy}}\big(\bar{\theta}^+\big). \tag{29}$$

Proof. By the definition of the differentiable post-update map in (28), the teacher parameters affect the outer objective only through $\bar{\theta}^+$. Applying the chain rule to $J_{\mathrm{out}}(\phi) = J_{\mathrm{phy}}(\bar{\theta}^+(\phi))$ gives (29). Hence, the meta-gradient combines the sensitivity of the post-update policy to the teacher parameters with the gradient of the trusted physical objective. This relation characterizes the objective-alignment mechanism under the differentiable surrogate, but does not by itself guarantee improvement of the practical policy update. □

Remark 1. The Meta Reward Teacher reduces the mismatch between corrupted rewards and the physical cooperative maneuvering objective. It corrects the task-level optimization direction during centralized training, while deployment uses the learned decentralized policies.

B. Action-Value Geometry Regularization via AMS

The Meta Reward Teacher improves learning-objective consistency, while local action-value evaluation remains sensitive to irregular variations in the continuous joint action space. This issue is particularly relevant for Ackermann vehicles because small variations in speed and steering commands may jointly influence curvature evolution and inter-vehicle coordination. Let the joint Ackermann action be

$$\boldsymbol{a} = [v_1, \delta_1, \ldots, v_N, \delta_N]^{\mathsf{T}} \in \mathcal{A}_{\mathrm{Ack}} \subset \mathbb{R}^{d_a}, \qquad d_a = 2N. \tag{30}$$

For an executed action $\boldsymbol{a}_e$, feasible symmetric neighbors are constructed as

$$\boldsymbol{a}_j^{\pm} = \Pi_{\mathrm{Ack}}(\boldsymbol{a}_e \pm \epsilon \boldsymbol{d}_j), \qquad j = 1, \ldots, K, \tag{31}$$

where Π_{Ack} enforces the speed, steering, and pairwise Ackermann feasibility constraints. The action directions satisfy $\boldsymbol{d}_i^{\mathsf{T}} \boldsymbol{d}_j = \delta_{ij}$. The neighborhood value is

$$\overline{Q}(\boldsymbol{s}, \boldsymbol{a}_e) = \frac{1}{2K} \sum_{j=1}^{K} \left[Q(\boldsymbol{s}, \boldsymbol{a}_j^{+}) + Q(\boldsymbol{s}, \boldsymbol{a}_j^{-}) \right]. \tag{32}$$

Assumption 2 (Local action-value regularity). For a fixed state $\boldsymbol{s}$, the action-value function $Q(\boldsymbol{s}, \cdot)$ is four times continuously differentiable in the considered neighborhood, with bounded second- and fourth-order derivatives. The feasibility projection is locally inactive or preserves the symmetric action pairs to the order required by the expansion. The direction ensemble is isotropic in expectation, satisfying

$$\mathbb{E}\left[\frac{1}{K} \sum_{j=1}^{K} \boldsymbol{d}_j \boldsymbol{d}_j^{\mathsf{T}} \right] = \frac{1}{d_a} I_{d_a}. \tag{33}$$

Proposition 1 (Implicit Local Action-Value Regularization). Under Assumption 2, the expected neighborhood value satisfies

$$\mathbb{E}\left[\overline{Q}(\boldsymbol{s}, \boldsymbol{a}_e)\right] = Q(\boldsymbol{s}, \boldsymbol{a}_e) + \frac{\epsilon^2}{2d_a} \Delta_{\boldsymbol{a}} Q(\boldsymbol{s}, \boldsymbol{a}_e) + O(\epsilon^4). \tag{34}$$

Proof. Expand $Q(\boldsymbol{s}, \boldsymbol{a}_e \pm \epsilon \boldsymbol{d}_j)$ around $\boldsymbol{a}_e$. The positive and negative samples cancel the first-order terms. Averaging the second-order terms and applying the isotropy relation in (33) gives the Hessian-trace term in (34). The fourth-order remainder is bounded under the stated smoothness assumption. Therefore, the symmetric neighborhood expansion yields the $O(\epsilon^4)$ remainder in (34). This result characterizes the local action-value regularization induced by neighborhood averaging under the stated smoothness, sampling, and feasibility conditions. □

Remark 2. AMS smooths local action-value geometry and improves the reliability of local policy updates; the task-level reward objective remains defined by the Meta Reward Teacher. In the implemented update, the twin critics provide the local value estimates and the corresponding bounded auxiliary advantage is combined with the HAPPO advantage as

$$A_{\mathrm{total}} = \mathrm{Norm}(A_{\mathrm{HAPPO}} + \lambda_k A_{\mathrm{AMS}}). \tag{35}$$

The Meta Reward Teacher and AMS operate at different learning levels: the former determines the physically meaningful optimization direction, whereas the latter regularizes how nearby feasible actions are evaluated during policy improvement.

C. Conditional Tracking Boundedness of Learned Ackermann Platoon

After the learning process converges to a feasible policy, the closed-loop tracking behavior of the resulting Ackermann platoon is characterized under bounded hybrid disturbances. Define the scaled tracking error vector and the aggregate attack disturbance as

$$\begin{aligned} \boldsymbol{e}_k &= S_e^{-1} \operatorname{col}(\boldsymbol{e}_{p,k}, \boldsymbol{e}_{s,k}, \boldsymbol{e}_{\psi,k}, \boldsymbol{e}_{v,k}), \\ S_e &= \operatorname{diag}(s_p I, s_s I, s_\psi I, s_v I), \end{aligned} \tag{36}$$

and

$$\boldsymbol{w}_k = \operatorname{col}(\boldsymbol{w}_{\mathrm{DoS},k}, \boldsymbol{w}_{\mathrm{FDI},k}, \boldsymbol{w}_{\mathrm{ext},k}), \qquad \|\boldsymbol{w}_k\| \le \bar{w} < \infty. \tag{37}$$

Assumption 3 (bounded hybrid attack). The DoS holding error, FDI perturbation, and external disturbance remain bounded in the considered operating domain, so that the bound in (37) holds.

Assumption 4 (local regularity of the learned closed loop). The learned closed-loop map is locally regular and admits a positive definite Lyapunov function $V(\boldsymbol{e})$ with constants $c_1, c_2, \alpha, c_w > 0$ such that

$$\begin{aligned} c_1 \|\boldsymbol{e}_k\|^2 \le V(\boldsymbol{e}_k) \le c_2 \|\boldsymbol{e}_k\|^2, \\ V(\boldsymbol{e}_{k+1}) - V(\boldsymbol{e}_k) \le -\alpha V(\boldsymbol{e}_k) + c_w \|\boldsymbol{w}_k\|^2. \end{aligned} \tag{38}$$

The learned closed-loop error dynamics are written locally as

$$\boldsymbol{e}_{k+1} = f_{\mathrm{cl}}(\boldsymbol{e}_k, \boldsymbol{w}_k), \qquad f_{\mathrm{cl}}(\boldsymbol{0}, \boldsymbol{0}) = \boldsymbol{0}. \tag{39}$$

The following theorem characterizes the conditional tracking boundedness of the learned Meta-AMS policy within the considered operating domain.

Theorem 2 (Conditional Tracking Boundedness). Under Assumptions 3–4, every trajectory that remains in the local operating domain satisfies

$$\limsup_{k \to \infty} \|\boldsymbol{e}_k\| \le \Delta_{\mathrm{UB}} := \sqrt{\frac{c_w}{\alpha c_1}}\, \bar{w}. \tag{40}$$

Proof. From (38) and Assumption 3, $V(\boldsymbol{e}_{k+1}) \le (1 - \alpha) V(\boldsymbol{e}_k) + c_w \bar{w}^2$. Iterating this recursion gives a geometric transient and the limiting bound $c_w \bar{w}^2 / \alpha$ for V. Using the lower quadratic bound $c_1 \|\boldsymbol{e}_k\|^2 \le V(\boldsymbol{e}_k)$ yields (40).

Remark 3. This analysis characterizes the conditional tracking performance of the learned Meta-AMS policy within the verified operating region and under the stated bounded-disturbance and local-regularity assumptions. The result is scoped to policies and attack budgets satisfying these conditions.

V. Simulation Protocol

A. Environment and Warm Start

The experiments evaluate the learning performance of Meta-AMS under progressive hybrid DoS–FDI attacks and the cooperative maneuvering capability of the five-vehicle Ackermann platoon. All experiments are conducted in NVIDIA Isaac Sim/Isaac Lab using a physics-based planar environment [15]. The vehicle interface enforces front-wheel steering, and each vehicle follows the common spatial lane-change reference while coordinating with its predecessor.

The policy is initialized from a behavior-cloning (BC) checkpoint trained on simulator-executed expert trajectories. Fixed behavior-cloning and reference-policy KL retention penalties are used during attack adaptation; the progressive expert-exit schedule is not activated in the reported experiments. The attack levels are evaluated using three fixed environment/attack seed pairs: $(42, 3407)$, $(3407, 9917)$, and $(777, 17011)$. A checkpoint is promoted to the next attack stage only when both the clean-task and current attacked-task completion criteria are satisfied.

The verified comparison uses 32 training environments, 16 evaluation environments, and a 40-update budget at each attack stage.

B. Attack Ladder and Metrics

The nine evaluated levels are *off*, *light*, *easy-0*, *easy-a*, *easy-b*, *med-a*, *med-b*, *hard-a*, and *hard-b*. A checkpoint is promoted only if the clean gate and the current attacked gate pass. We report:

1) whole-platoon completion rate over the evaluation environments;
2) the minimum completion rate among the three evaluation seeds;
3) episode return;
4) collision rate, road-boundary violation rate, and road-termination rate;
5) qualitative evidence from the common spatial trajectory, top-down path, and physical front-steering traces.

VI. Results

A. Spatial and Ackermann Evidence

The cooperative maneuver experiment evaluates whether the learned policy can coordinate longitudinal and lateral motion under the Ackermann steering constraint. Fig. 2 presents the shared spatial reference and the sequential trajectories of the five vehicles, while Fig. 3 shows the corresponding tracking errors and front-steering responses. The results demonstrate that the simulated platoon can follow the common lane-change path using front-wheel steering, with the rear wheels providing driving motion only. This experiment validates the maneuvering capability under the specified vehicle model and evaluation conditions.

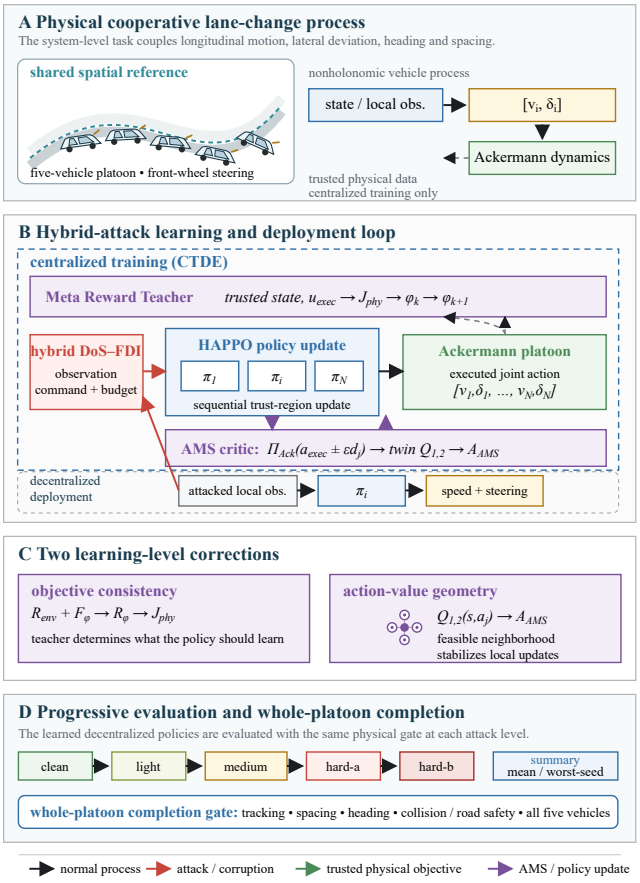


Fig. 1. Meta-AMS architecture. The reward teacher uses privileged physical information only during centralized training. AMS evaluates local feasible action neighborhoods and contributes a bounded auxiliary advantage to the HAPPO update.

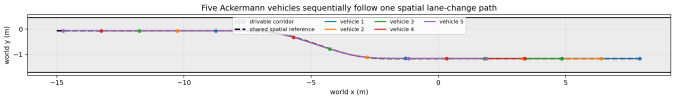


Fig. 2. Five Ackermann vehicles sequentially following one shared spatial lane-change path in the real-trajectory evidence.

B. Meta-AMS Attack Ladder

Table I summarizes the whole-platoon completion performance under the progressive hybrid-attack protocol. The Meta-AMS policy passes the completion criterion at all stages through hard-a, achieving a completion rate of 1.00. At hard-b, the completion rate is 0.75, and the worst-seed result is also 0.75. No collisions, road-boundary violations, or road terminations are recorded in the reported evaluation matrix. The evaluation therefore characterizes one verified training chain under the specified attack levels.

TABLE I
META-AMS PERFORMANCE UNDER THE PROGRESSIVE HYBRID-ATTACK PROTOCOL.

Level	off	light	easy-0	easy-a	easy-b	med-a	med-b	hard-a	hard-b
Completion mean	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.75
Worst seed	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.75
Episode return	73782	71928	69744	68278	66634	65198	64470	61414	55884
Collision rate	0	0	0	0	0	0	0	0	0
Road violation	0	0	0	0	0	0	0	0	0
Road termination	0	0	0	0	0	0	0	0	0

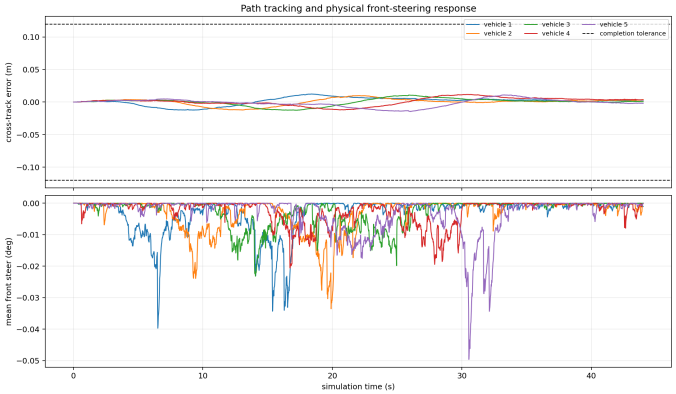


Fig. 3. Cross-track error and physical front-steering response for the Ackermann warm-start trajectory.

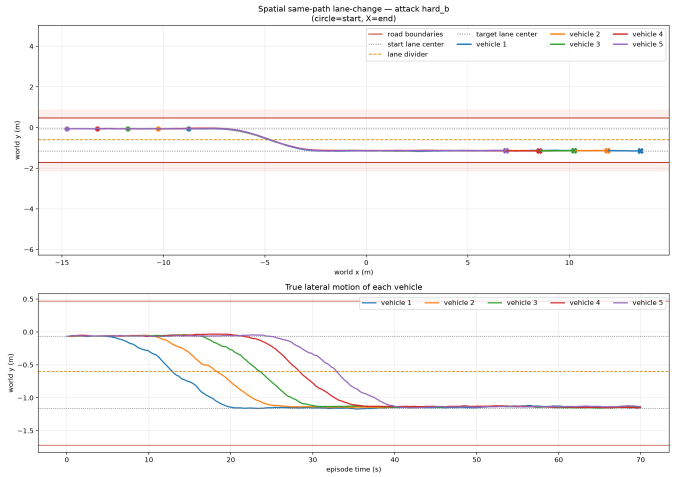


Fig. 4. Meta-AMS hard-b top-down trajectory. Circles denote starts and crosses denote final positions; all five vehicles reach the target lane while remaining within the road corridor.



Fig. 5. Meta-AMS hard-b reward, path/gap/heading errors, and attack/safety signals. The recorded road-exceedance trace remains zero.

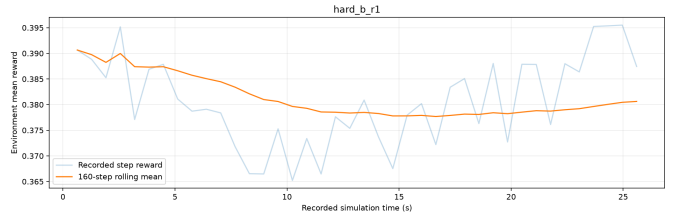


Fig. 6. Recorded Meta-AMS hard-b stage reward curve and rolling mean.

C. Component Analysis of Meta-AMS

Table II provides a diagnostic comparison of the learning components rather than a complete causal ablation. The Meta Reward Teacher addresses attack-induced reward misalignment through trusted physical-performance feedback, whereas AMS supplies auxiliary local action-value information through feasible action neighborhoods. The HAPPO + AMS-only branch stopped at the light stage, and its hard-b entry corresponds to the last safe behavior-cloning parent rather than a trained AMS-only checkpoint. This entry therefore does not provide an independent evaluation of AMS. The combined Meta-AMS configuration achieves the highest completion rate in the diagnostic table. The current results do not isolate the individual causal contribution of each component.

D. Comparison With MARL Methods

Table III compares the whole-platoon completion performance of Meta-AMS with the selected MARL baselines at hard-b. Meta-AMS achieves a mean completion rate of 0.7500 and a worst-seed completion rate of 0.7500, compared with 0.6250 and 0.5625 for MAPPO and 0.5833 and 0.5000 for HAPPO, respectively. The compared methods exhibit different levels of performance degradation under the considered attack condition. Under this protocol, Meta-AMS has the highest mean and worst-seed completion rates among the compared methods. Independent training runs are needed to evaluate the statistical significance and generality of the observed differences.

TABLE II
DIAGNOSTIC COMPONENT ANALYSIS OF META-AMS AT HARD-B. THE AMS-ONLY ROW CORRESPONDS TO THE LAST SAFE BEHAVIOR-CLONING PARENT RATHER THAN A TRAINED AMS-ONLY CHECKPOINT.

Method	Completion mean	Worst seed	Training status
HAPPO base shared	0.5208	0.50	passed hard-b passed
HAPPO + Meta	0.5208	0.50	hard-b stopped
HAPPO + AMS only	0.0417	0.00	at light passed
HAPPO + Meta + AMS	0.7500	0.75	hard-b

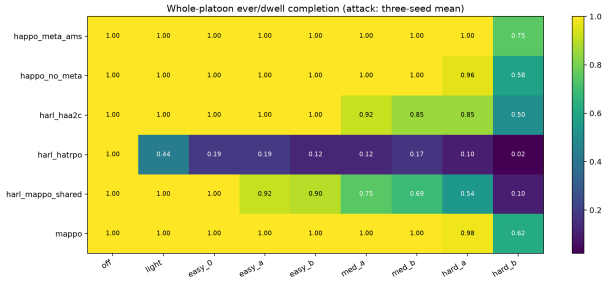


Fig. 7. Whole-platoon mean completion over the complete attack ladder for the six-algorithm comparison.

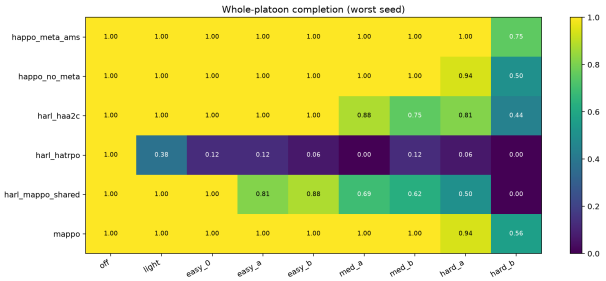


Fig. 8. Worst-seed whole-platoon completion over the attack ladder. The worst-seed view exposes the hard-b degradation more clearly than the mean alone.

TABLE III
HARD-B COMPARISON FROM THE VERIFIED SIX-ALGORITHM MATRIX.

Method	Completion mean	Worst seed	Gate
MAPPO	0.6250	0.5625	pass
HAPPO (no Meta)	0.5833	0.5000	pass
Meta-AMS	0.7500	0.7500	pass
HARL-HAA2C	0.5000	0.4375	fail
HARL-MAPPO shared	0.1042	0.0000	fail
HARL-HATRPO	0.0208	0.0000	fail

E. Interpretation

The matrix shows three findings. First, the full Meta-AMS policy is the only reported method with a 0.7500 hard-b whole-platoon completion rate and a 0.7500 worst-seed rate. Second, collision and road termination remain zero in the reported matrix. Third, the diagnostic comparison shows the highest completion rate for the combined Meta Reward Teacher and AMS configuration; separate causal contributions require independent component checkpoints.

VII. DISCUSSION AND LIMITATIONS

A. What Meta-AMS Adds to Meta-Gradient Reward Shaping

The MGRS formulation addresses attack-induced reward mismatch by exploiting trusted physical information in the upper-level update [8]. Meta-AMS keeps that task-level separation and adds a local one. The teacher selects a physically meaningful learning direction, whereas AMS evaluates nearby speed–steering actions for value and feasibility. For a nonholonomic platoon, this division links gap-error reduction to local steering perturbations that preserve curvature and front-wheel consistency. These two perspectives provide complementary directions for improving the security of safety-critical autonomous decision-making systems.

B. Experimental Evidence Boundary

The reported evidence comes from one verified Meta-AMS training chain evaluated using three fixed environment/attack seed pairs under the specified progressive hybrid-attack protocol. The behavior-cloning initialization uses simulator-executed expert trajectories, and the study is simulation-based. Independent training runs and controlled component ablations are needed to quantify generality and separate the contributions of the Meta Reward Teacher and AMS.

C. Theory Boundary

The local ultimate-boundedness proposition characterizes the learned closed loop within the verified operating region under bounded attacks and local regularity. The simulation gates test shared-path completion, gap maintenance, dwell, collision, and road-boundary constraints. Future studies can relate logged trajectory data to the boundedness constants and measurable action-value variation.

VIII. CONCLUSION

This paper presented Meta-Guided Action-Manifold Smoothing for secure five-vehicle Ackermann platooning under hybrid DoS–FDI attacks. The method combines a trusted physics-aware Meta Reward Teacher with a feasible, symmetric, twin-critic action neighborhood. The vehicle interface enforces front-wheel steering and rear-wheel drive, while the curriculum begins from simulator-executed expert trajectories and promotes checkpoints through strict complete-platoon gates. The verified Meta-AMS training chain reaches 0.7500 whole-platoon completion at hard-b with zero collision and road-termination rates in the reported matrix, and the spatial/top-down evidence shows all five vehicles following the same path and completing the lane change. Future independent training runs and controlled component ablations can quantify statistical variation and component-level effects.

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